

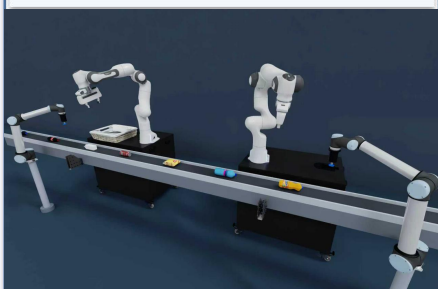


1 Inputs

Task Instruction
"I need a bottle of Cola, Mirinda and Milk in the Basket."

+

Scene



Visual Observation $o(t)$

Capability Sets $\{C_i\}$



2

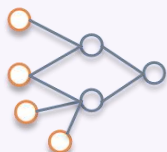
SEMANTIC LAYER

Command Dispatch

Franka_left PICK milk
 Suction_left PICK cola
 ⋮
 Franka_right GO_TO pos_1

Shared Semantic State

S_{ν}^{sem}



authoritative · versioned

Affordance Role

- Objects in which region?
- Reachable by which arm?
- Any obstruction?
- Any scene changed?

Planning Role

- Subtask decomposition
- Capability assignment
- Skill match
- Construct dependency

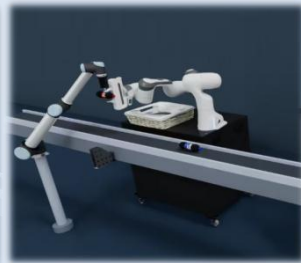
Verification Role

- Execution abstract
- Subtask succeeded?
- Any events occurred?
- Which subtask affected?



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PHYSICAL LAYER



Execution Evidence

Event
 · task update
 · disturbance
 · failure

Event Occurs



"No, use a can of Cola instead of Mirinda."

→ online execution flow

- - - semantic event re-entry